Voice-enabled Assistants of the Operator 4.0 in the Social Smart Factory: prospective role and challenges for an advanced Human-Machine Interaction

Abstract

Greater cognitive task load and growing shortage of high-skilled labor call for new smart interactions between the cyber-physical production system (CPPS) and the Operator 4.0. Voice-enabled assistants (VA) gives users intuitive access to a plethora of information and knowledge. However, their implementation still finds limited attention in industrial contexts. This article sheds light on the prospective adoption and acceptance of VAs to assist the Operator 4.0 during industrial production processes within the Social Smart Factory. Leveraging on quality-driven engineering of human-machine interaction systems and on the prototyping of a VA for a CNC milling machine, insights and challenges are discussed.

Keywords

Industry 4.0; Smart Factory; Operator 4.0; Human-Machine Interaction; Voice-enabled Assistant

1. Introduction

Decreasing lot sizes and higher product variety are among the key features of the 4th industrial revolution [1]. Given the greater cognitive task load required to workers who need to master a variety of work tasks and tools [2] and the growing shortage of skilled labor [3], there is a growing demand for assistance technology [4]. This calls for new kinds of smart interactions between the cyber-physical production system (CPPS) and the "new" Operator 4.0 that support decision-making or the execution of a task [5]. Augmented Reality (AR) has widely demonstrated the benefits of a digitalized knowledge management for assisting operators to complete specific tasks [6], e.g. by showing visual information directly in the technician's field of view [7–9]. Today, assistance systems are now converging with another promising innovation – vocal-based interaction – that gives workers intuitive access to a plethora of information. However, despite research on voice-enabled assistants (VA) is not new, their implementation in industrial contexts still finds limited attention.

This article sheds light on the prospective adoption and acceptance of VAs to assist the Operator 4.0 within the Social Smart Factory. Leveraging on quality-driven engineering of human-machine interaction systems and on the prototyping of a VA for a CNC milling machine, insights and challenges are discussed.

2. Voice-enabled Assistants: trend and prospective role in the Social Smart Factory

Voice-enabled Assistants (VAs), such as Amazon's Alexa or Google Assistant, are drastically rising in popularity [10]. In 2017, seoClarity reported that the most frequently occurring terms in user queries to home-installed VAs are "How" and "What", showing that VA's users often ask for informational content and procedural support [11]. In the manufacturing domain, some authors [12] envision smart VAs for human-robot communication. However, despite research has recently started to maximize users' cognitive efficiency in assembly tasks [13] and to ascertain the VA's benefits on productivity and process quality [14], they still present many shortcomings that weaken their perceived usefulness and acceptance in industry.

Prospectively, every single autonomous and cooperative building block of a CPPS – i.e. the holon [15] – may be enabled with voice and intelligence, thus giving birth to the Social Smart Factory (Figure 1). A social network of VAs associated to every single holon is the key of the Social Smart Factory, in conjunction with IoT, CPPS and Digital Twin technologies. The Operator 4.0 can interact via voice responses with the VA of a networked manufacturing resource (e.g. asking a machine instructions for a particular work program), of a work order (e.g. asking which is the next step to accomplish), of the environment (e.g. asking a room information about lighting or temperature), of a particular product (e.g. asking about its components) or of the workers themselves (e.g. asking about health information). It is the concept of a Human CPPS [16] in which the Operator 4.0 interacts with the digital factory through intuitive interfaces – i.e. spoken dialogue.

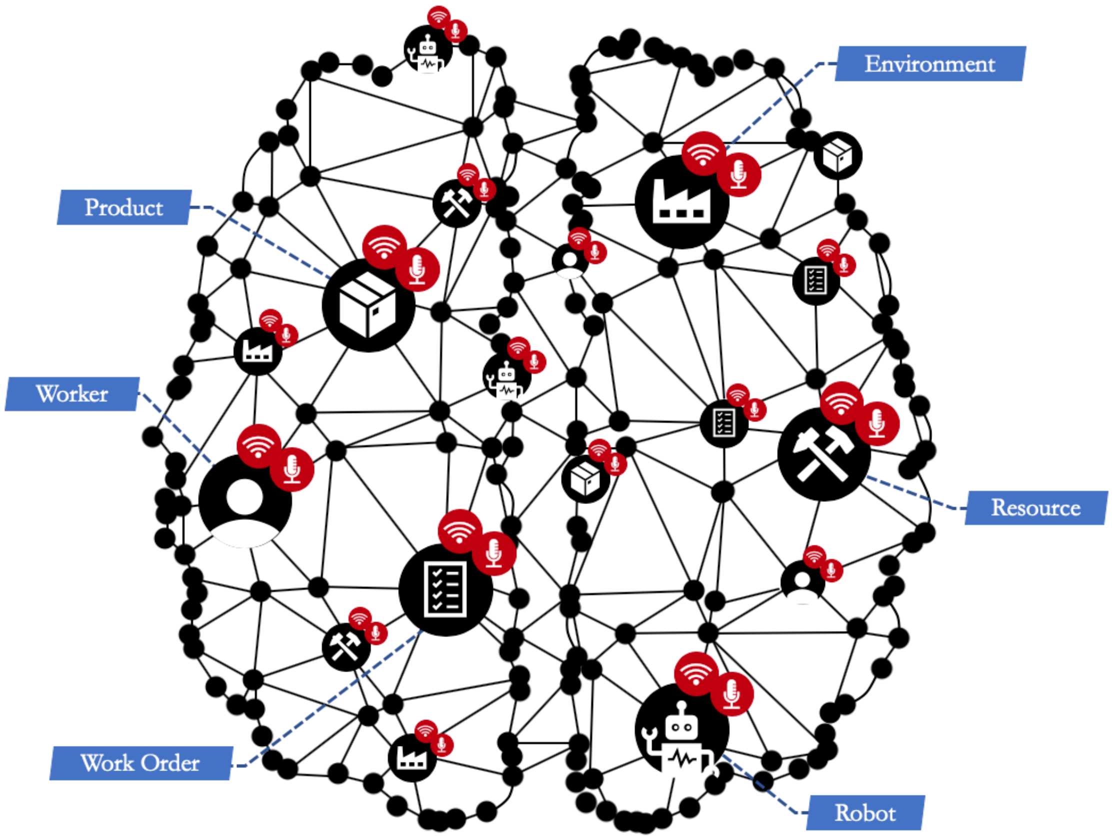

Figure 1. Social Smart Factory: enabling manufacturing holons with voice and intelligence

3. Voice-enabled Assistants for the Social Smart Factory: Quality Engineering Requirements

Perceived usefulness and ease-of-use represent the key metrics according to which a VA satisfies the needs of various stakeholders, and thus provide value to the company. Regardless of the VA's degree of intelligence or hardware where it is installed, quality engineering considerations were derived from the "SQuaRE" framework described in the ISO/IEC 25010 standard to provide guidance on, and requirements for the implementation of VAs in a Smart Factory (Figure 2).

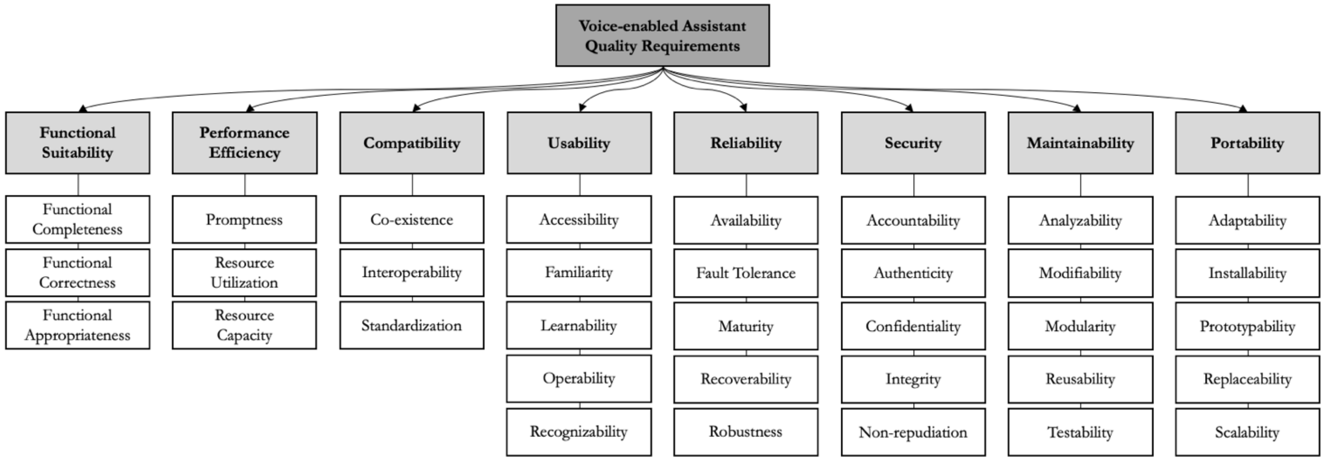

Figure 2. Quality engineering requirements and challenges for VAs in the Social Smart Factory (adapted from ISO/IEC 25010)

4. Prototyping a Voice-enabled Assistant for the Operator 4.0: a Discussion on Design Choices and Open Challenges

Thanks to the application of the SQuaRE framework during the prototyping, installation and testing of a VA for a CNC milling machine (Figure 3), insights about design choices and open challenges are discussed. The end-users are maintenance operators, machine operators and novice workers who can use the VA for a twofold objective:

i. to detect a machine failure and receive step-by-step assistance through the operation restoring procedure;
ii. to conduct the machine correctly and safely.

Identifying the end-users is a crucial step to align the VA to their actual needs and provide value (e.g. faster learning, easier information retrieval, higher safety, less stress). Therefore, a comprehensive skillset for an industrial machine's VA, i.e. a set of built-in capabilities the Operator 4.0 can engage with, has been developed in cooperation with a group of 8 industry experts. It includes capabilities to access information about components, corrective and preventive maintenance, consumables and spare parts, programs, control panel and signals, and safety guidelines.

Figure 3. VA Prototype: on-screen response while VA is speaking (top-left), step-by-step VA-assisted procedure (bottom-left), rich content – e.g. video, image – proposed by the VA with specific info about machine component while VA is speaking (top- and bottom-right)

4.1. How the user interacts with the VA

In-field testing demonstrated that a large set of users' open-ended utterances can be correctly interpreted by a keyword-based mechanism associated to a 'priming' conversational design [17]. This design scheme successfully addresses also discoverability and memorability issues. Indeed, although voice interaction makes the users quickly familiar with the VA, it would not be intuitive to understand at first use or after a period of not using it what and how to ask things. A discrete speech recognition approach (i.e. an interval of no user input triggers turn-taking) has been designed to enable a pleasing dialogue that is not limited to two conversational turns. During the experimentation with the VA, users have been asking first information about a machine alarm and its solution, in the middle of which they asked information about machine components, safety procedures and spare parts. This approach is also suitable for people with auditory/speaking impairments that can use a screen to type their question and see the response (Figure 3). Dialogues were recorded for accountability purposes in order to trace back who was the operator who accomplished a specific task, e.g. preventive maintenance, thus automating a procedure that is currently done by the companies on paper. An interaction example between the VA and the operator is given in Figure 4.

4.2. How the VA interacts with the user

The VA provides messages in the form of a prompt, which requires further instructions from the user. Both industry experts and scholars agree on the need to design a voice persona that evokes a distinct tone and personality in order to create trust and enhance the acceptance rate by different groups of users. Despite user's voice profiling may significantly improve the accurateness and prevent from unauthorized access to the application (privacy considerations are still needed though), the developed recognition method has received large consensus with the experts. It is capable to make repairs, repeat the message or request an explicit confirmation when its confidence is low. Using pre-set answers and a high-quality synthesized voice to talk at a slow speaking pace represented accepted design choices by the experts.

4.1. Data entry and management: the application back-end

The data entry process resulted to be very time-consuming as most of conceptual and procedural knowledge as well as domain grammar/dictionary is either tacit or not processable (available in old manuals). A user-friendly web app has been developed to fill out an ontology-based knowledge representation that can be adapted to every manufacturing holon thanks to a flexible tree structure and keyword labeling mechanism. Access authorization, versioning and accountability are necessary to prevent from data corruption and from potential serious risks. Instead, fault tolerance and data recovery plans in case of failures of the VA were not implemented in the prototype, despite they are considered necessary by the industry experts. What is interesting is that data entry is a non-core process that can be outsourced, e.g. to machine or equipment manufacturers. Smart machines and equipment may indeed come with built-in VA system and later integrated in the Social Smart Factory.

4.2. Installation and maintainability

A distributed, service-oriented and cloud-based architecture embodies the principles of the Social Smart Factory. The multi-platform application prototype has been developed for smartphone, tablets and HMDs, but also headset microphones. Such technologies reached today a significant maturity and a reliable speech-to-text even in noisy manufacturing environments, but further analysis and case studies are needed on this topic. The modular nature of this holonic architecture facilitates installation and maintainability but also poses several challenges. Interoperability standards and data security requirements are therefore an urgent need to support design choices for data exchange between the VA and any other system/application.

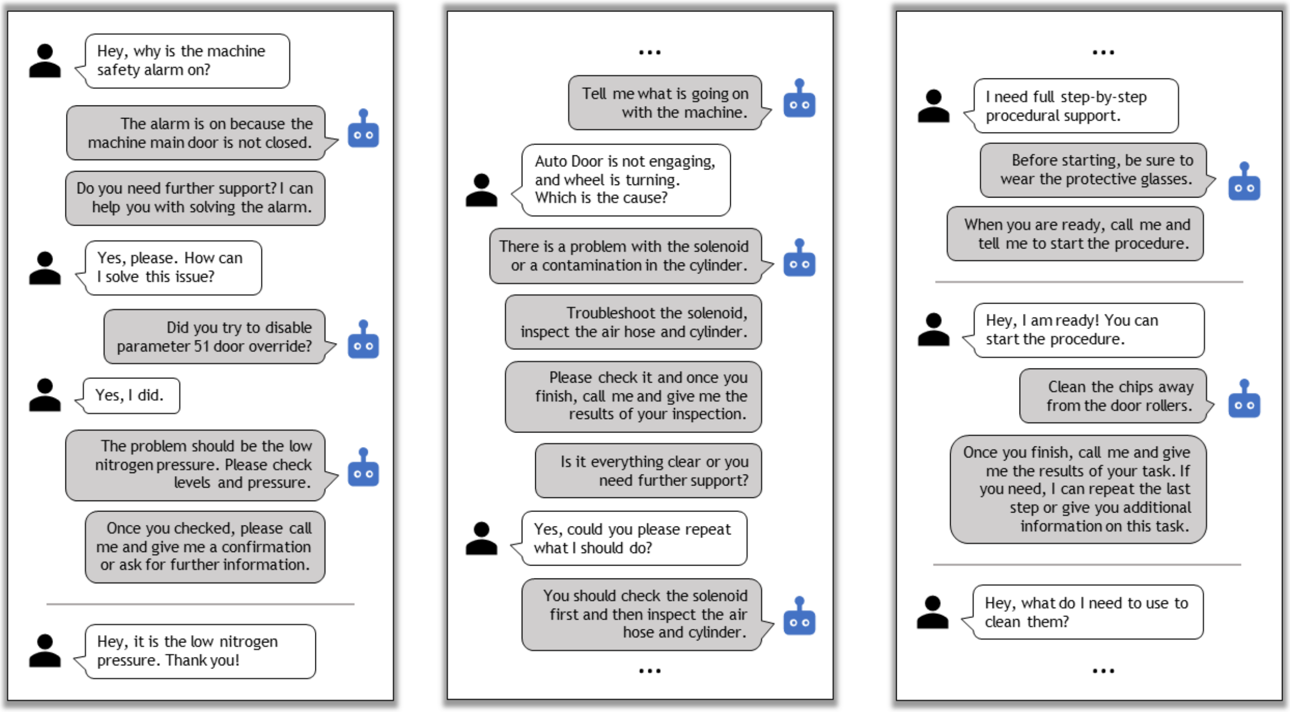

Figure 4. Interaction example between the operator and the VA. From left to right: asking for information about a machine alarm, for corrective action in case of machine failure, for step-by-step assistance during task execution.

5. Conclusions

Although further research is needed for a more precise voice recognition and conversational intelligence, technology is now mature enough to investigate the concept of a Social Smart Factory and the impact of VAs in the manufacturing floor. Long setup times, the lack of VA fast-prototyping tools and resistance to change by skeptical employees still discourage companies from adopting VAs, but they may have a disruptive effect if proper trajectories of technological development are engaged in by academia and business models are devised by forward-looking companies to make VAs beneficial in practice. Future work will focus on the conversational design and knowledge representation for VAs, as well as their integration with the Digital Twin.

Acknowledgements

This work is supported by the SO4SIMS project (Smart Operators 4.0 based on Simulation for Industry and Manufacturing Systems) funded by the Italian Ministry of Education, Universities and Research MIUR (Project PRIN – 2017FW8BB4).

Declaration of Competing Interest

We the authors declare that there is no conflict of interest to disclose.

References

[1] Tran NH, Park HS, Nguyen QV, Hoang TD. Development of a smart cyber-physical manufacturing system in the Industry 4.0 context. Appl Sci 2019;9:3325. https://doi.org/10.3390/app9163325.

[2] Guerin C, Rauffet P, Chauvin C, Martin E. Toward production operator 4.0: Modelling Human-Machine Cooperation in Industry 4.0 with Cognitive Work Analysis. IFAC-PapersOnLine, vol. 52, Elsevier B.V.; 2019, p. 73–8. https://doi.org/10.1016/j.ifacol.2019.12.111.

[3] World Economic Forum. The Future of Jobs Employment, Skills and Workforce Strategy for the Fourth Industrial Revolution. 2016. https://doi.org/10.1177/1946756712473437.

[4] Ras E, Wild F, Stahl C, Baudet A. Bridging the skills gap of workers in industry 4.0 by human performance augmentation tools - Challenges and roadmap. ACM Int Conf Proceeding Ser, 2017, p. 428–32. https://doi.org/10.1145/3056540.3076192.

[5] acatech – Deutsche Akademie der Technikwissenschaften; Kompetenzentwicklungsstudie Industrie 4.0 – Erste Ergebnisse und Schlussfolgerungen. München: 2016.

[6] Zhang J, Ong SK, Nee AYC. RFID-assisted assembly guidance system in an augmented reality environment. Int J Prod Res 2011;49:3919–38. https://doi.org/10.1080/00207543.2010.492802.

[7] Blattgerste J, Renner P, Strenge B, Pfeiffer T. In-situ instructions exceed side-by-side instructions in augmented reality assisted assembly. ACM Int Conf Proceeding Ser, 2018, p. 133–40. https://doi.org/10.1145/3197768.3197778.

[8] Webel S, Bockholt U, Engelke T, Gavish N, Olbrich M, Preusche C. An augmented reality training platform for assembly and maintenance skills. Rob Auton Syst 2013;61:398–403.

https://doi.org/10.1016/j.robot.2012.09.013.

[9] Sand O, Büttner S, Paelke V, Röcker C. smARt.Assembly – Projection-based augmented reality for supporting assembly workers. Lect Notes Comput Sci (including Subser Lect Notes Artif Intell Lect Notes Bioinformatics), vol. 9740, 2016, p. 643–52. https://doi.org/10.1007/978-3-319-39907-2_61.

[10] Statista Research Department. Number of digital voice assistants in use worldwide 2019-2023 2020. https://www.statista.com/statistics/973815/worldwide-digital-voice-assistant-in-use/#statisticContainer (accessed March 2, 2020).

[11] SeoClarity. The Next Generation of Search: Voice. 2017.

[12] Ghofrani J, Reichelt D. Using voice assistants as HMI for robots in smart production systems. CEUR Workshop Proc, vol. 2339, 2019, p. 62–5.

[13] Wang Z, Bai X, Zhang S, He W, Zhang X, Zhang L, et al. Information-level AR instruction: a novel assembly guidance information representation assisting user cognition. Int J Adv Manuf Technol 2020;106:603–26. https://doi.org/10.1007/s00170-019-04538-9.

[14] Longo F, Nicoletti L, Padovano A. Ubiquitous knowledge empowers the Smart Factory: The impacts of a Service-oriented Digital Twin on enterprises' performance. Annu Rev Control 2019;47:221–36. https://doi.org/10.1016/j.arcontrol.2019.01.001.

[15] Babiceanu RF, Chen FF. Development and applications of holonic manufacturing systems: A survey. J Intell Manuf 2006;17:111–31. https://doi.org/10.1007/s10845-005-5516-y.

[16] Zolotová I, Papcun P, Kajáti E, Miškuf M, Mocnej J. Smart and cognitive solutions for Operator 4.0: Laboratory H-CPPS case studies. Comput Ind Eng 2020;139. https://doi.org/10.1016/j.cie.2018.10.032.

[17] Reitter D, Moore JD, Keller F. Priming of Syntactic Rules in Task-Oriented Dialogue and Spontaneous Conversation. Proc 28th Annu Conf Cogn Sci Soc, 2006, p. 685–90.